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PhD position

Theory and Algorithms for Discrete Generative Modeling

📍 Institut Camille Jordan · INSA Lyon · France

We invite applications for a fully funded PhD position at the Institut Camille Jordan, INSA Lyon (Lyon, France), with earliest starting date october 2026.

Project description

Discrete generative modeling is essential for learning complex distributions over structured data such as text, graphs, and biological sequences. Developing and understanding models for discrete spaces is therefore a key challenge in modern machine learning, with broad implications for both theory and applications.

The PhD project focuses on the theoretical foundations and algorithmic design of discrete generative models, situated at the intersection of stochastic processes, probability theory, and machine learning.

The PhD candidate will investigate continuous-time Markov chains (CTMCs) and jump-diffusion processes for generative tasks. A core objective is to overcome the “factorization trap” present in current discrete flow matching models by exploring alternative combinatorial and geometric structures on the space of probability measures. This will require investigating combinatorial and geometric structures on probability spaces that go beyond classical Wasserstein geometry. The candidate will work both on the theoretical aspects and on the implementation of novel architectures, including hybrid jump-diffusion flows in token embedding spaces.

Candidate profile

We are looking for a highly motivated candidate with:

- A Master’s degree in Mathematics, Computer Science, or a closely related field.
- A strong background in probability theory and stochastic processes is a plus.
- Interest in the mathematics of machine learning and generative modeling.
- Programming experience; knowledge of Python and frameworks such as PyTorch is a plus.

How to apply

Interested candidates should send the following documents by **August 15**:

- CV (including contact details of references)
- academic transcripts
- a statement describing interest and relevant experience

If you are are interested or have any question, please contact:
Christian Wald (✉ christian.wald@insa-lyon.fr)

Relevant literature

The project will build upon recent advances in generative modeling and stochastic processes. Some key references include:

- C. Wald, G. Steidl, *Flow Matching: Markov Kernels, Stochastic Processes and Transport Plans* (2025).
- P. Holderrieth et al., *Generator Matching: Generative modeling with arbitrary Markov processes* (2024).
- Y. Lipman et al., *Flow Matching Guide and Code* (2024).
- M. A. Peletier, R. Rossi, G. Savaré, O. Tse, *Jump processes as generalized gradient flows* (2022).
- G. Kornhardt, J. Chemseddine, C. Wald, G. Steidl, *Self-Aware Markov Models for Discrete Reasoning* (2026).
- A. Campbell et al., *A Continuous Time Framework for Discrete Denoising Models* (2022).
- J. Austin et al., *Structured Denoising Diffusion Models in Discrete State-Spaces* (2021).